



Luiz Carazolli

Mathematician
M.Sc. candidate, Unicamp

CONTACT

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LANGUAGES

Portuguese — native
English — C2 (2017)
Spanish — intermediate
Italian — intermediate

TECHNICAL SKILLS

Programming
Python, R, Mathematica, $\text{\LaTeX}$

Scientific computing
NumPy, pandas, GUDHI

Machine learning
PyTorch, scikit-learn, XGBoost, LightGBM, SHAP

Research
Mathematical reasoning and writing; computational experiments; model evaluation; teaching.

SELECTED HONORS

Gold medals / 2020
National Science Olympiad;
Brazilian Astronomy Olympiad;
Unicamp Mathematics Olympiad.

Silver medal / 2020
Brazilian Physics Olympiad.

Profile

I’m a CAPES-funded master’s candidate in Mathematics at Unicamp, where I also earned my bachelor’s degree. My research connects algebraic geometry and symplectic topology, with a focus on orbifold Gromov–Witten theory and quantum cohomology. I also have applied research experience in AI and deep learning, including natural language processing and deep neural networks. Building on this background, I’m developing interests in topological data analysis and the mathematical foundations of deep learning. I’m seeking a position in the Anthropic Fellows Program.

Education

M.Sc. in Mathematics 2025–present
University of Campinas (Unicamp)
Research: Orbifold Quantum Cohomology. Advisors: Prof. Dr. Lino Grama and Dr. Leonardo Cavenaghi. Funding: CAPES.

B.Sc. in Mathematics 2021–2024
University of Campinas (Unicamp)

Publications

A quantum connection approach to rationality 2026
Luiz Carazolli, Leonardo F. Cavenaghi, Ludmil Katzarkov, Maxim Kontsevich, and Pedro Antonio Muniz Martins. Preprint. [arXiv:2610.06669](#).

Research Experience

Research Intern — Artificial Intelligence & Deep Learning 2022
GoToData | MCTI-funded internship
Applied research spanning machine learning, natural language processing, and deep neural networks. Focus areas included deep learning approaches to language processing and neural-network methods for data science.

Special Lagrangian Manifolds via Examples 2024
Research internship | ICMS, Bulgarian Academy of Sciences
One-month research visit studying special Lagrangian submanifolds through examples in complex Euclidean space and Calabi–Yau threefolds. Supervision: Prof. Dr. Ludmil Katzarkov and Prof. Dr. Lino Grama. Funding: FAPESP–BEPE, grant 2024/07883-0.

Introduction to Symplectic Geometry 2023–2024
Undergraduate research | University of Campinas
Studied foundations of symplectic geometry, coadjoint orbits of $\text{SL}(2, \mathbb{R})$, and an introduction to generalized complex geometry. Supervisor: Prof. Dr. Lino Grama. Funding: FAPESP, grant 2023/07412-4.

Differential Forms and the Geometry of Surfaces 2022–2023
Undergraduate research | University of Campinas
Studied differential forms and Cartan’s moving-frame method, culminating in a proof of the Gauss–Bonnet theorem for surfaces. Supervisor: Prof. Dr. Lino Grama. Funding: CNPq.

Selected Projects

The shape of a photograph

2026

Topological data analysis | Machine learning | Computer vision

Exploratory study of whether the topology of grayscale intensity landscapes can distinguish real photographs from AI-generated images.

- Built a Python pipeline using GUDHI to compute cubical persistent homology and extract 16 scalar topological descriptors from CIFAKE images.
- Compared six tabular classifiers with a PyTorch residual CNN operating on two-channel sampled persistence surfaces; used SHAP to explore feature attributions.
- The saved CNN achieved 74.70% accuracy and 0.8404 ROC-AUC on a 4,000-image internal holdout from a 20,000-image pool. The experiment re-splits CIFAKE's test partition and does not use its official train/test protocol.

[Project page](#), [technical report](#), [code](#), and [reproducible workspace](#).

Teaching

Teaching Assistant — General Topology

2026

IMECC, University of Campinas | Graduate-level course

Teaching Assistant — Introduction to Analysis

2025

IMECC, University of Campinas | Undergraduate-level course

Selected Talks

Introduction to Orbifolds

2024

Seminar.

The Gauss–Bonnet Theorem via the Moving-Frame Method

2023

34th Brazilian Mathematics Colloquium.

Differential Forms and the Geometry of Surfaces

2023

Unicamp Undergraduate Research Congress.

Selected Training

Foundations of Business Strategy — University of Virginia, 2025.

Differentiable Manifolds — UFMG, 2024.

Introduction to Banach Spaces — IMPA, 2023.

General Topology — UFMG, 2023.